

#### Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

##### CODE

```
import java.util.Scanner;

class Employee {
    int empId;
    String empName;
    double basicSalary;

    void getEmployeeDetails(int empId, String empName, double basicSalary) {
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Salary s = new Salary();

        int id = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double salary = sc.nextDouble();

        s.getEmployeeDetails(id, name, salary);
        s.calculateSalary();
        s.displayDetails();
    }
}

class Salary extends Employee {
    double hra, da, netSalary;

    void calculateSalary() {
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        netSalary = basicSalary + hra + da;
    }

    void displayDetails() {
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Net Salary: " + netSalary);
    }
}
```

##### INPUT

101  
Indhu  
50000

##### OUTPUT

Employee ID: 101  
Employee Name: Indhu  
Basic Salary: 50000.0  
HRA: 10000.0  
DA: 5000.0  
Net Salary: 65000.0

### Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Student {

    int rollNo;
    String name;
    double marks;

    Student() {
        rollNo = 101;
        name = "Rahul";
        marks = 85.5;
    }

    Student(int rollNo, String name, double marks) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }

    void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Student s1 = new Student();
        s1.displayDetails();

        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();

        Student s2 = new Student(rollNo, name, marks);
        s2.displayDetails();
    }
}
```

INPUT

102  
Indhu  
95.5

OUTPUT

Roll No: 101  
Name: Rahul  
Marks: 85.5  
Roll No: 102  
Name: Indhu  
Marks: 95.5

## Q2. Classes and Objects-Initialize Object Data Using Setter Methods

Score: 10.00    TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Student {

    private int rollNo;
    private String name;
    private double marks;

    void setRollNo(int rollNo) {
        this.rollNo = rollNo;
    }

    void setName(String name) {
        this.name = name;
    }

    void setMarks(double marks) {
        this.marks = marks;
    }

    void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Student s = new Student();

        s.setRollNo(sc.nextInt());
        sc.nextLine();
        s.setName(sc.nextLine());
        s.setMarks(sc.nextDouble());

        s.displayDetails();
    }
}
```

INPUT

```
101
Indhu
95.5
```

OUTPUT

```
Roll No: 101
Name: Indhu
Marks: 95.5
```

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5 / 5 submitted

**Q1. Program based on Classes and Objects - Initializaing Data member using Objects directly**

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Student {

    int rollNo;
    String name;
    double marks;

    void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Student s = new Student();

        s.rollNo = Integer.parseInt(sc.nextLine());
        s.name = sc.nextLine();
        s.marks = Double.parseDouble(sc.nextLine());

        s.displayDetails();

        sc.close();
    }
}
```

INPUT

181  
Rahul  
85.5

OUTPUT

Roll No: 181  
Name: Rahul  
Marks: 85.5

## Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

### CODE

```
import java.util.Scanner;

class Employee {
    int empId;
    String empName;

    void getEmployeeDetails(int empId, String empName) {
        this.empId = empId;
        this.empName = empName;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        GrossSalary g = new GrossSalary();

        int id = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double basicSalary = sc.nextDouble();

        g.getEmployeeDetails(id, name);
        g.getBasicSalary(basicSalary);
        g.calculateSalary();
        g.displayDetails();
    }
}

class Salary extends Employee {
    double basicSalary;

    void getBasicSalary(double basicSalary) {
        this.basicSalary = basicSalary;
    }
}

class GrossSalary extends Salary {
    double hra, da, grossSalary;

    void calculateSalary() {
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        grossSalary = basicSalary + hra + da;
    }

    void displayDetails() {
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Gross Salary: " + grossSalary);
    }
}
```

### INPUT

101  
Indhu  
50000

### OUTPUT

Employee ID: 101  
Employee Name: Indhu